

ECE 684 Natural Language Processing – Assignment 3

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Assumptions of the Noisy Channel Model:

1. **Single Error Per Word:** It is assumed that each word contains at most one error, meaning all errors are within an edit distance of one.
2. **Exclusion of Transpositions:** The model does not account for transposition errors, focusing solely on deletions, insertions, and substitutions.
3. **Vocabulary Limitations:** The vocabulary is restricted to the unigram frequency corpus provided by Peter Norvig. Words not included in this corpus are considered invalid.

Testing the code:

After building the code, I used two sets of inputs, and the outputs are as follows:

```
161 # test cases
162 print("Cases where the Noisy Channel model should work as expected")
163 print(f"input word: helo, corrected word: {correct('helo')}")
164 print(f"input word: definately, corrected word: {correct('definately')}")
165 print(f"input word: tee, corrected word: {correct('tee')}")
166 print(f"input word: tye, corrected word: {correct('tye')}")
167 |
168 print("-----")
169 # test cases which give the wrong output
170 print("Cases where the Noisy Channel Model did not work as expected")
171 print(f"input word: recieve, corrected word: {correct('recieve')}") # the output should be "receive" but transposition is not considered in this model
172 print(f"input word: quizardacious, corrected word: {correct('quizardacious')}") # rare words that don't occur often. the correct spelling should be "quizzacious"
173
```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS Code

[Running] python -u "/Users/nrutachoudhari/Documents/MIDS/Fall 2024/ECE 684 Natural Language Processing/Noisy Channel Model - Spelling Correction 2009/tempCodeRunner"

Cases where the Noisy Channel model should work as expected
input word: helo, corrected word: hello
input word: definately, corrected word: definitely
input word: tee, corrected word: tree
input word: tye, corrected word: the

Cases where the Noisy Channel Model did not work as expected
input word: recieve, corrected word: recieved
input word: quizardacious, corrected word: quizardacious

[Done] exited with code=0 in 0.829 seconds

As shown in the screenshot, the output is as expected for the words where there is a substitution, addition or deletion of letters.

However, the model fails when there are cases added where there are transpositions and where the word is rare. In those cases, the output is the same as the misspelled word.